RESIDENTIAL ENERGY CONSUMPTION PREDICTION USING MACHINE LEARNING MODELS FOR SUSTAINABLE ENERGY MANAGEMENT

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Abstract

To achieve sustainable decision making in data-driven settings, you need accurate, robust and future-generalising predictive models. This paper deals with the sustainability issue of making predictive analytics, which are reliable, to support long-term operational and planning choices by adapting a data mining strategy.

The data of this research has been received in an open database, and it consists of records on transactions, having numeric, categorical, and time attributes. To measure the effects of a temporal attribute, two versions of the datasets were analysed: with date attributes and without them. A range of predictive modelling methods were utilized, such as Decision Tree, Neural Network, Gradient Boosting, HP Forest, and Ensemble model. Validation Average Squared Error (ASE) was used to assess model performance and provide fair comparison and allow generalisation. The findings show that HP Forest (T3) had the lowest validation ASE in both variants of data, and it has shown superior results than all other models. Date attributes made some models more efficient, but HP Forest proved to be highly robust irrespective of feature representation.

These results justify the creation of a sustainable decision support solution through the development of a predictive model that balances accuracy and generalisation. The model chosen has the potential to help organisations take informed long-term decisions and reduce uncertainty and resource efficiency.

Keywords—Data Mining, Predictive Analytics, Sustainable Decision Support, Ensemble Learning Techniques

I. INTRODUCTION

The increasing demand for electricity has bring concern related to environmental sustainability, energy efficiency and rising household costs. Residential buildings had brought a substantial portion of total energy consumption, where most of them come from appliance, lighting and air conditioning. energy usage wastage not only leads to higher electricity bills, but it also increases the carbon emissions that contribute to climate change. Thus, improvement in residential energy efficiency has become an important sustainability challenge for everyone.

The advancements in sensor technologies and smart home systems have enabled the collection of data on household energy usage and indoor environmental conditions. These data will provide an opportunity to apply data mining techniques to identify pattern and relationships that unable observed by traditional analysis methods. By using these data, predictive model can be developed to support informed decision making that aim to reduce unnecessary energy consumption while maintaining people's comfort.

This study uses a real-world residential energy consumption dataset containing appliance and lighting energy usage, indoor temperature, humidity measurements from multiple rooms, outdoor weather variables that obtained from nearby weather station in Belgium. The main objective of this research is to develop a sustainable decision-support solution capable by predict household energy consumption using data mining techniques.

The objectives of this study are:

1. To identify which environmental and weather factors affect the residential energy consumption.
2. To evaluate the effectiveness of ensemble models and neural networks in predicting household energy usage.
3. To demonstrate how predictive analytics can support sustainable energy management decisions.

II. LITERATURE REVIEW

This section is a literature review of available research on predicting energy consumption in homes using data-driven and machine learning methods. It summarises existing literature to find out commonly used modelling methods, compare them in their merits and shortcomings,

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and comment on the way the recent developments respond to increasing accessibility of high-resolution smart-home sensor data. The review also identifies gaps in current research, thus defining the scope and contribution of this study.

A. Existing work on Residential Energy Consumption Prediction

Energy consumption prediction of residential and building has been widely researched in literature, especially as the number of smart-meters and indoor environmental sensor datasets are becoming more popular and accessible. Linear regression and other statistical models were widely used in the early research because they are simple and easy to interpret [1], [2]. Such models generally associate energy consumption with a limited set of explanatory variables, commonly including outdoor temperature and time-related factors. Nevertheless, thorough review articles have consistently indicated that linear models lack the ability to simulate the non-linear dynamic nature of residential energy use, which is shaped by complicated interactions amid environmental conditions and occupant behaviour [3]-[5].

To address these shortcomings, later studies moved towards more machine learning approaches, such as decision trees, support vector machines, ensemble learning frameworks, and neural networks [3], [4], [6], [7]. The decision tree-based techniques are also employed as part of broader machine learning pipelines in recent energy-related forecasting research to capture nonlinear interactions among sensor, temporal, and contextual variables in addition to providing rule-based interpretability that can be used to gain insight into consumption behaviour and model decisions [6], [7]. These methods are more appropriate to high dimensional, sensor rich data and have been used widely in residential and building energy prediction studies [3]-[5].

B. Modelling Techniques in Prior Studies

There are also clear differences in methodology between earlier studies in terms of predictive accuracy, predictive robustness, and interpretability. Linear and statistical models are often utilized as standard methods because they are a transparent method and have a low cost in computation [1], [2]. Nevertheless, it is reported in few studies and reviews that such models usually fail to perform well when applied to real-world household data which is nonlinear and whose behaviour is variable [3], [5].

Decision trees, as an intermediate approach between linear regression and more complex ensemble-based models, have been applied to energy forecasting problems as rule-based interpretable models that can integrate context-dependent and sensor-dependent information (i.e. time periods and environmental/occupancy-related signals) to assist in decision logic to solve prediction problems [7]. Recent studies do however also indicate that single-tree techniques may be quite weak, motivating ensemble extensions (e.g. bagging/boosting of trees) to help in generalisation in complex energy datasets [6].

Ensemble learning, especially the Random Forest and Gradient Boosting are highly used in overcoming these challenges. Random Forest is a method published by Breiman [8], which integrates multiple decision trees to minimise variation and enhance generalisation, and is useful with high-dimensional data with correlated variables. Across multiple residential energy forecasting studies, tree-based ensemble methods often outperform simpler statistical baselines, especially when both indoor environmental variables and outdoor weather features are included [9]-[11]. Still, ensemble techniques have low interpretability and need to be hyperparameter tuned to prevent overfitting [3], [5].

Neural network methods additionally expand the predictive power by modelling difficult nonlinear interactions of environmental and time features. Initial studies by Hippert et al. showed that artificial neural networks could predict short term loads more efficiently than traditional statistical techniques [12]. Recent papers utilize deep neural networks and recurrent models with residential energy data and achieve higher predictive accuracy especially with time-series forecasting [13]-[15]. However, these models often require a lot of pre-processing, feature scaling, and parameter optimisation and their complexity can limit transparency, which restricts transparency [3], [4].

Despite the extensive reported use of ensemble models and neural networks as high-performing methods, numerous studies test them separately in varying experimental settings. Direct comparisons using consistent preprocessing pipelines, feature sets, and evaluation metrics remain relatively limited in literature.

C. Algorithmic Handling of Temporal Features

The incorporation of time-related data is also widely known to be an important factor in residential energy prediction because the patterns of consumption are cyclical in nature. However, according to the recent literature the way these features include are depends on whether they are considered as raw integers or transformed values, as it have differently effective depending on the algorithm used.

Previous research suggests that algorithms that use decision trees such as Random Forest and Gradient Boosting are especially efficient in working with raw temporal features without significant pre-processing. It has been demonstrated that the decision tree structures can be successfully applied to the discrete time variables by splitting a dataset in the Day and Night periods to choose the best forecasting strategies [7]. This is in line with the results which show that tree-based models are especially

good in the representation of non-linear relationships using IF-THEN rule and hence are capable of interpreting categorical or integer-based time characteristics directly [6]. It has also been emphasized that categorical variables are more easily modelled, which is an added benefit of the ensemble-based algorithms in constructing energy applications because they do not need the strict normalization or encoding that is common to the other algorithms [10].

On the other hand, many standard Artificial Neural Networks (ANNs) have challenges on working with raw, integer-based temporal features. Another major weakness determined in the previous studies is that the representation of cyclic variables as linear numerical variables creates discontinuities in the data that are not natural [16]. For example, a neural network will perceive an Hour 23 and Hour 0 as numerically distant numbers, even though they are temporally close to each other. Such discontinuity may be a barrier to gradient-based optimization schemes in neural networks. In practice, however, deep learning designs are not as robust as they need to be as they may need extra feature engineering to effectively capture time as a continuous and cyclic domain [16].

D. Strengths and Limitations of Current Studies

Altogether, the available literature confirms that most advanced machine learning models, especially the use of ensemble learning and the use of neural networks, are more accurate and robust than conventional statistical models in predicting residential energy consumption because they can trace nonlinear trends and feature interactions [3][5]. The addition of indoor environmental factors (temperature, humidity) has been reported to greatly improve predictive accuracy over those models that used only outdoor weather data [10][17].

Nonetheless, there are still a few constraints in previous literature. First, many studies emphasise predictive accuracy while providing limited discussion on how model outputs can be translated into actionable insights for sustainable energy management [18]. Second, decisions related to preprocessing of data, including extreme energy usage values, skewed target distributions, and temporal feature extraction, are not always explained enough, even though they each might have a significant influence on model performance and reproducibility [3][5]. Third, the preprocessing pipelines and evaluation strategies vary across studies making it difficult to directly compare the reported outcomes [3][4][5].

Despite multiple studies suggesting that ensemble models and neural networks have strong predictive performance, the literature gives little information on how these models could be compared systematically within a single preprocessing and evaluation framework. Furthermore, numerous papers highlight accuracy measures without clearly showing how the outputs of the models could be used to assist in the real-life decision-making in residential management of energy with sustainability in mind. Consequently, there is still a necessity of studies that systematically compare advanced modelling methods within a unified experimental structure and directly connects the findings of predictive methods to decision-support objectives in residential energy sustainability [3][4][5][18].

As a result, there is a need of studies that systematically compares advanced modelling methods within a unified experimental structure and directly connects the findings of predictive methods to decision-support objectives in residential energy sustainability.

E. Contribution of Study

Building on the literature review, this work adds value through a controlled comparison between the models of ensemble learning and neural networks on a single and well-documented preprocessing pipeline and evaluation metrics. The problems of nonlinearity and high-dimensionality in the literature of the previous works are directly addressed in the present work through the use of sensor-rich residential dataset that measures the appliance level of energy consumption and offers detailed indoor and outdoor variables of the environment. In addition, the paper also focuses on exploring environmental factor influence interpretation alongside predictive accuracy, which encourages utilizing data-based models as practical decision-support systems to manage residential energy sustainably.

III. METHODOLOGY

A. Problem and Data Understanding

This study addresses the challenge of high residential electricity consumption that increased household costs and carbon emissions. The goal of this study is to develop a data-driven decision-support solution that can predict household energy usage and identify which environmental factors may affect it. By getting an accurate prediction, this help residents and energy stakeholders to optimize energy usage, reduce wastage and improve long-term sustainability.

This research uses the Appliances Energy Prediction dataset obtained from UCI Machine Learning Repository. The dataset represents real-world smart-home energy monitoring data that collected from a residential building in Belgium. It records 10-minute intervals within 4.5 months. The indoor temperature and humidity date were collected using a wireless sensor network that had installed in multiple room of a residential building. Besides that, the appliance and lighting energy usage represented the household energy consumption. In addition, the weather also be considered and obtained from nearest airport weather station.

This dataset has contained more than 10000 records and includes variable such as energy usage (Wh), temperature (℃), humidity (%), wind speed (m/s) and atmospheric pressure. Furthermore, two random variables were also included to support model validation and to identify non-predictive attributes. The decision-support objective is to develop predictive models that can estimate household energy consumption and provide insights for sustainable energy management.

The following table shows the variable with their description, role and data type:

Variable Name	Description	Role	Data Type
date	The record date and time	Feature	Date
Appliances	energy use (Wh)	Target	Integer
lights	energy use of light fixtures in the house (Wh)	Feature	Integer
T1	Temperature in kitchen area (℃)	Feature	Continuous
RH_1	Humidity in kitchen area (%)	Feature	Continuous
T2	Temperature in living room area (℃)	Feature	Continuous
RH_2	Humidity in living room area (%)	Feature	Continuous
T3	Temperature in laundry room area (℃)	Feature	Continuous
RH_3	Humidity in laundry room area (%)	Feature	Continuous
T4	Temperature in office room (℃)	Feature	Continuous
RH_4	Humidity in office room (%)	Feature	Continuous
T5	Temperature in bathroom (℃)	Feature	Continuous
RH_5	Humidity in bathroom (%)	Feature	Continuous
T6	Temperature outside the building (north side)(℃)	Feature	Continuous
RH_6	Humidity outside the building (north side) (%)	Feature	Continuous
T7	Temperature in ironing room(℃)	Feature	Continuous
RH_7	Humidity in ironing room (%)	Feature	Continuous
T8	Temperature in teenager room 2(℃)	Feature	Continuous
RH_8	Humidity in teenager room 2 (%)	Feature	Continuous
T9	Temperature in parent's room (℃)	Feature	Continuous
RH_9	Humidity in parent's room (%)	Feature	Continuous
T_out	Temperature outside (℃)	Feature	Continuous
Press_mm_hg	Pressure (mm Hg)	Feature	Continuous
RH_out	Humidity outside (%)	Feature	Continuous
Windspeed	Wind speed (m/s)	Feature	Continuous
Visibility	Visibility (km)	Feature	Continuous
Tdewpoint	Tdewpoint (℃)	Feature	Continuous
rv1	Random Variable 1, nondimensional	Feature	Continuous
rv2	Random Variable 2, nondimensional	Feature	Continuous

B. Data Preprocessing

Data Quality and Cleaning

Missing Value & Duplicate

The data were initially evaluated to provide a solid baseline on which to base the modelling. An initial test with Microsoft Excel functions ensured that no values or doubled records were missing. In order to strictly prove this, the StatExplore node of SAS enterprise miner was used. The findings (Figure 2) supported the preliminary results, showing that there were no missing values in all observations, which means that the imputation methods were not necessary.

Noise Removal

A number of non-predictors were found and eliminated to enhance the efficiency of the model. The variables rv1 and rv2 were dropped because the documentation of the dataset states explicitly that they are random variables that are added to test regression models and that they do not hold any information that is useful to predict energy. Moreover, the empty variables (Var1) created during the file import procedure were assigned with the value of ID since it is just a unique identifier of the observations and lacks predictive patterns when raw.

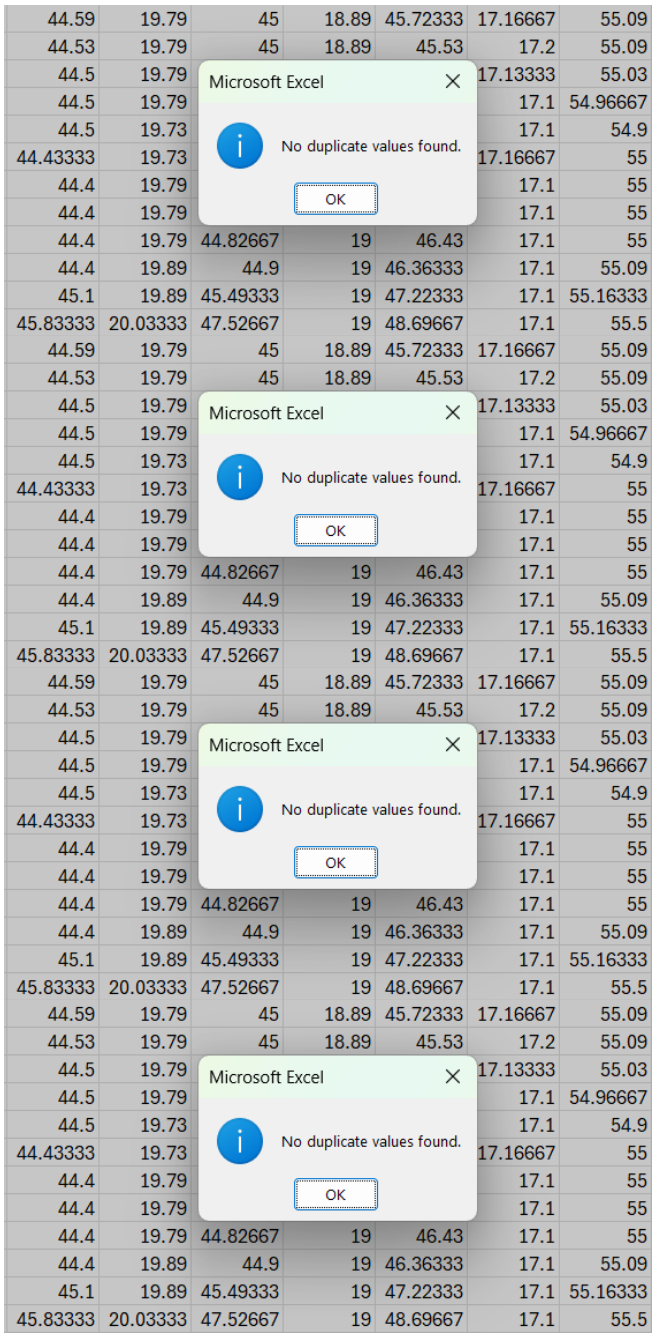


Figure 1

Variable	Role	Mean	Standard Deviation	Non Missing	Missing
Press_mm_hg	INPUT	755.5226	7.399441	19735	0
RH_1	INPUT	40.25974	3.979299	19735	0
RH_2	INPUT	40.42042	4.069813	19735	0
RH_3	INPUT	39.2425	3.254576	19735	0
RH_4	INPUT	39.0269	4.341321	19735	0
RH_5	INPUT	50.94928	9.022034	19735	0
RH_6	INPUT	54.60908	31.14981	19735	0
RH_7	INPUT	35.3882	5.114208	19735	0
RH_8	INPUT	42.93617	5.224361	19735	0
RH_9	INPUT	41.5524	4.151497	19735	0
RH_out	INPUT	79.75042	14.90109	19735	0
T1	INPUT	21.68657	1.606066	19735	0
T2	INPUT	20.34122	2.192974	19735	0
T3	INPUT	22.26761	2.006111	19735	0
T4	INPUT	20.85533	2.042884	19735	0
T5	INPUT	19.59211	1.844623	19735	0
T6	INPUT	7.910939	6.090347	19735	0
T7	INPUT	20.26711	2.109993	19735	0
T8	INPUT	22.02911	1.956162	19735	0
T9	INPUT	19.48583	2.014712	19735	0
T_out	INPUT	7.41258	5.318464	19735	0
Tdewpoint	INPUT	3.760995	4.195248	19735	0
Visibility	INPUT	38.33083	11.79472	19735	0
Windspeed	INPUT	4.039752	2.451221	19735	0
lights	INPUT	3.801875	7.935988	19735	0
Appliances	TARGET	97.69496	102.5249	19735	0

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Figure 2

Figure 3

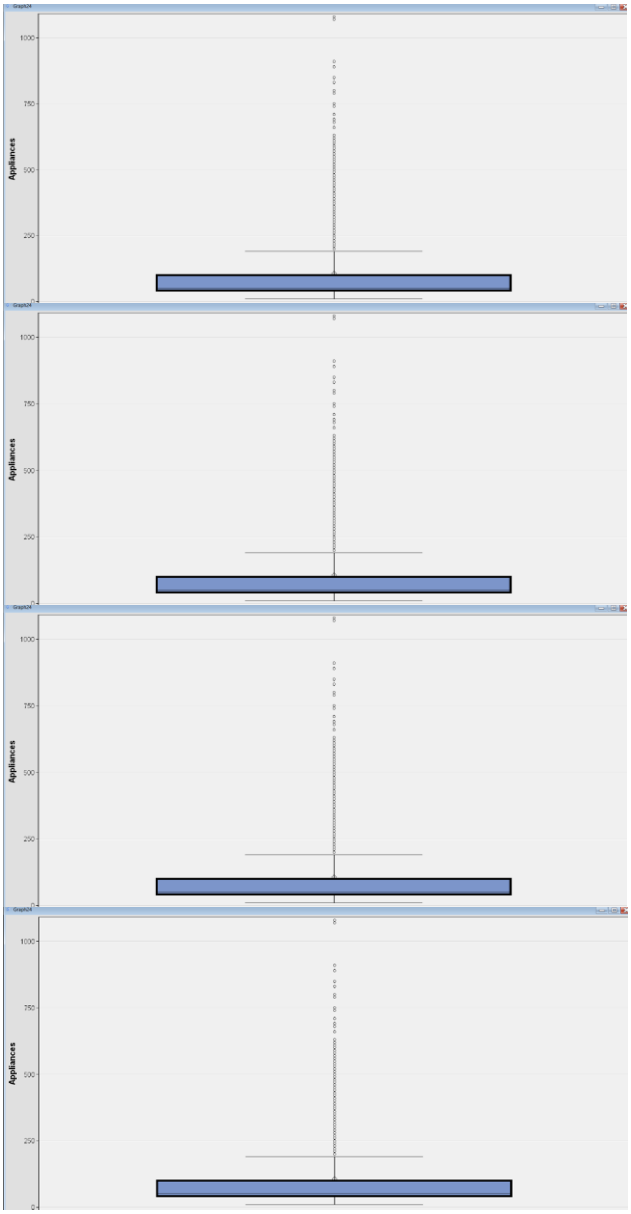


Figure 4

Next, an outlier analysis was conducted on the target variable, Appliances, using a box plot (figure 4) and a histogram (figure 5). From the box plot, it revealed significant outliers in the upper range that extends up to 1000 Wh, while the histogram confirmed a highly right-skewed distribution. However, these outliers were retained despite the statistical deviation, this is because in the context of building energy sustainability, these spikes represent legitimate peak-load events such as operating high-wattage appliances rather than sensor errors. Thus, removing them would prevent the models to learn on predict peak energy demand, which is critical for the decision-support objective.

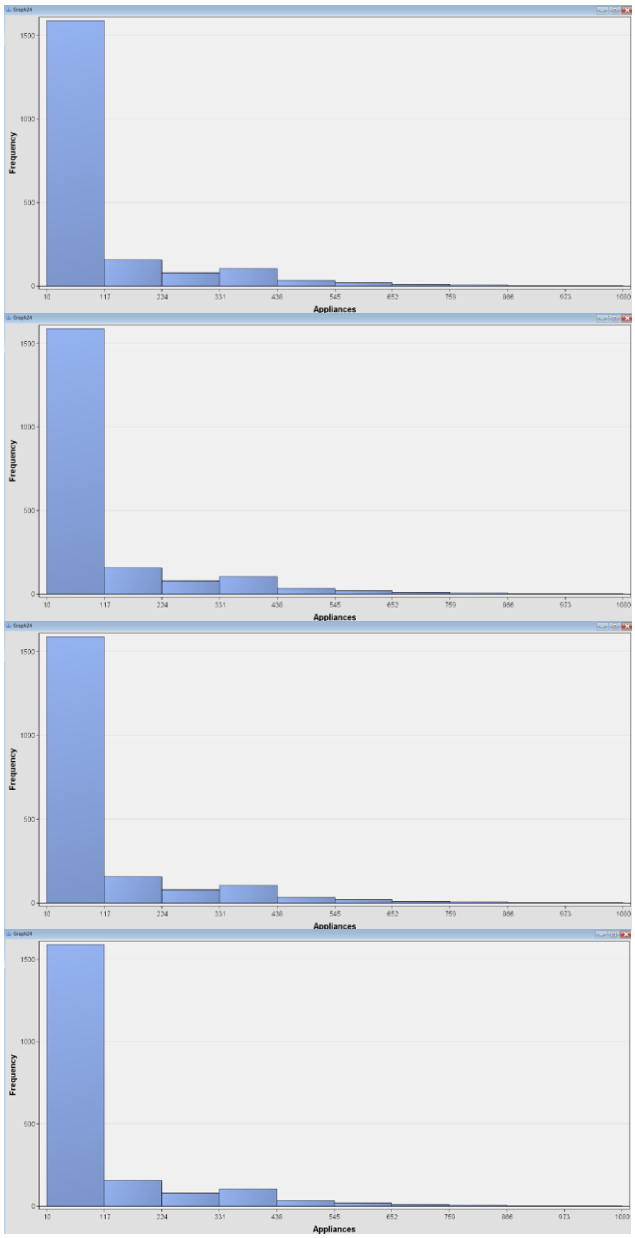


Figure 5

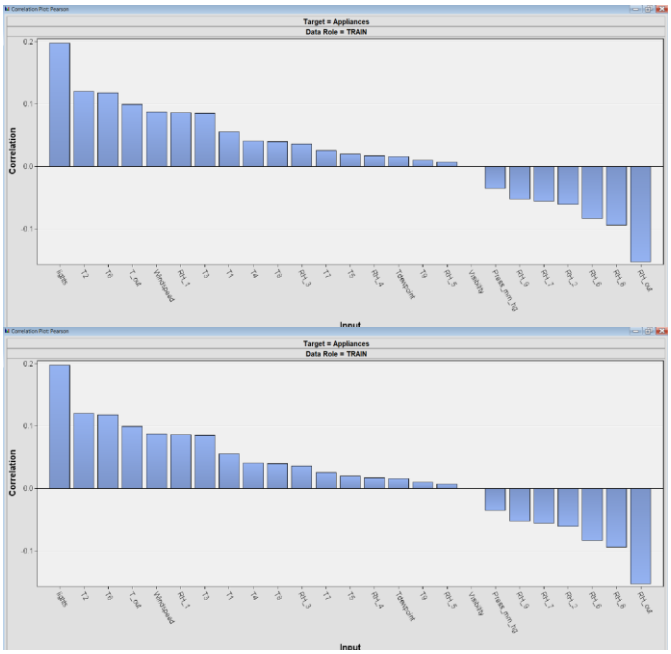
Feature Engineering and Algorithm-Specific Selection

Timestamp Data

In order to extract the cyclical trends observed at the stage of Data Understanding, the first step involved breaking down the timestamp information into three clear features; HourofDay, DayofWeek and Month. Nevertheless, it was noted that various mathematical architectures do not handle temporal data in the same way, so a feature utility analysis was incorporated into the modeling pipeline. This assessment aimed to compare the effects of explicit integer time characteristics on model stability- in other words, to test whether explicit integer time characteristics were informative splitting rules in tree-based models or introduced numerical noise in continuous neural networks. As a result, a protocol featuring a algorithm-specific feature selection was established: the final input sets were not optimized on a global scale but optimized on a model type basis. Using this strategy, only explicit temporal features would be retained by algorithms which proved to show a quantifiable improvement in performance over the initial sensitivity checks, and rejected by those which showed improvement with cyclical discontinuities.

Correlation Analysis

A Pearson correlation analysis from figure 6 was conducted to verify feature relevance. The analysis showed that lights had the strongest positive correlation (~0.2), while RH_out (Outside Humidity) had the strongest negative correlation. Significantly, all linear correlations were weak ($r < 0.2$). This weak linear dependency indicates that energy consumption is driven by complex, non-linear interactions, justifying the use of advanced algorithms like Neural Networks and Ensemble Trees over simple linear regression.



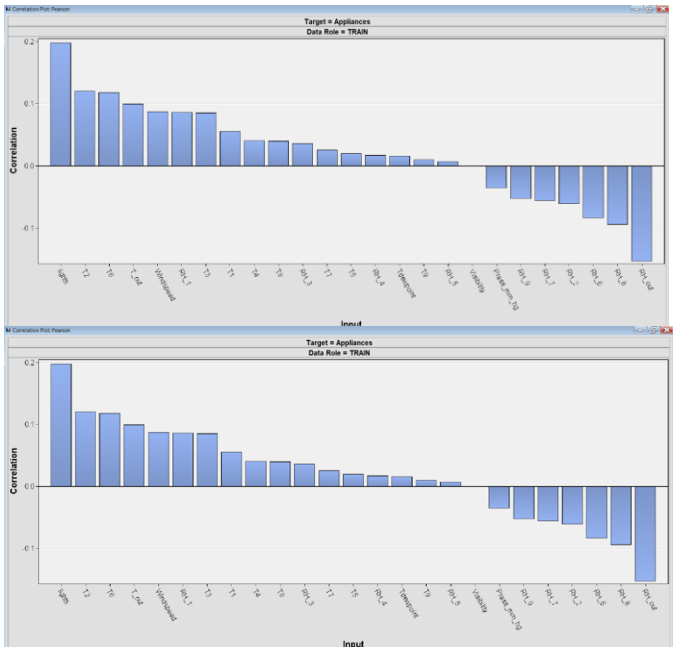


Figure 6

Data Transformation

In order to normalize the heterogeneous sensor data to be ready to model it, the Min-Max normalized technique was used with Transform Variables node. This normalized all the continuous input variables (including Temperature, Humidity and Windspeed) to a normalized range [0, 1] (Figure 7). Although tree-based algorithms typically do not depend on feature scaling, it was required in the Neural Network models. Neural networks are built on gradient descent optimization, and may fail to converge or become biased when input features have extremely different values (ex: Atmospheric Pressure of the order of 760 mmHg and Windspeed of the order of 5 m/s). Normalizing is needed to guarantee even or unbiased comparison of every model in a similar feature space.

Data Partitioning

A sensitivity analysis was performed to determine the optimal amount of training data needed to reach the optimal model and to assess the stability of the model. The data was divided into three different partitions ratios (Training: Validation): 70:30, 80:20 and 90:10. In this way, the study will be able to determine whether higher training data (90:10) that can be maximized is capable of improving the model in terms of capturing rare peak-load events, which is necessary to meet the sustainability objective..

C. Modeling Approaches

In this study, decision tree, gradient boosting, random forest, neural network and ensemble models is used to make the prediction model for household energy consumption. These modelling techniques were selected as they have the strong performance in regression task. Besides that, they able to capture complex relationships between variables and energy usage.

Firstly, a decision tree model will develop as a baseline predictive model that capture non-linear relationships between through hierarchical splitting rules. Following this, gradient boosting model is used to enhance the predictive accuracy of decision tree by correcting the prediction errors. This help to reduce bias and improve overall prediction performance. Both models will combine using standard ensemble node in SAS Enterprise miner to produce a single prediction that leverage the strengths of each model.

The other ensemble technique is random forest that use HP forest node. It aggregates prediction from multiple decision trees that help to improve robustness and reduce overfitting. Furthermore, Neural networks used to capture complex non-linear interaction between indoor environmental conditions.

In contrast to the tree-based models, the Neural Network used continuous sensor data only as time proxies to eliminate noise due to explicit integer variables. This method has worked in uncovering some hidden patterns in continuous data that might not be fit by conventional models. The neural network enabled extensive comparison of modelling strategies to facilitate the choice of the best model in sustainable energy decision-support.

D. Evaluation Strategy

The evaluation of models involved a test of the predictive performance of all models based on a uniform validation measure so as to make a fair comparison. The Validation Average Squared Error (VASE) was chosen as the primary criterion for ranking the models. VASE is used to determine the mean of the squared difference between the predicted and actual energy consumption values. Most importantly the Validation partition was employed instead of the Training partition to attempt to penalize the overfitting on the training data as strictly as possible. In addition to this, the model with minimum VASE was proclaimed the Champion Model.

Although VASE was used as the mathematical selection criteria, the root means square error (RMSE) were employed to explain the practical business meaning of the findings. In contrast to squared errors, these measures indicate the unit of prediction error, which is the original unit of the target variable (watt-hours) and can enable a concrete evaluation of the usefulness of the model in any automation application. Lastly, model stability was tested by looking at the training error versus the Validation error. A large split in which training error is much smaller than validation error had been observed as an indicator of overfitting. This was to make sure that the final model chosen would be reliable in its generalization to serve sustainable energy management in the house.

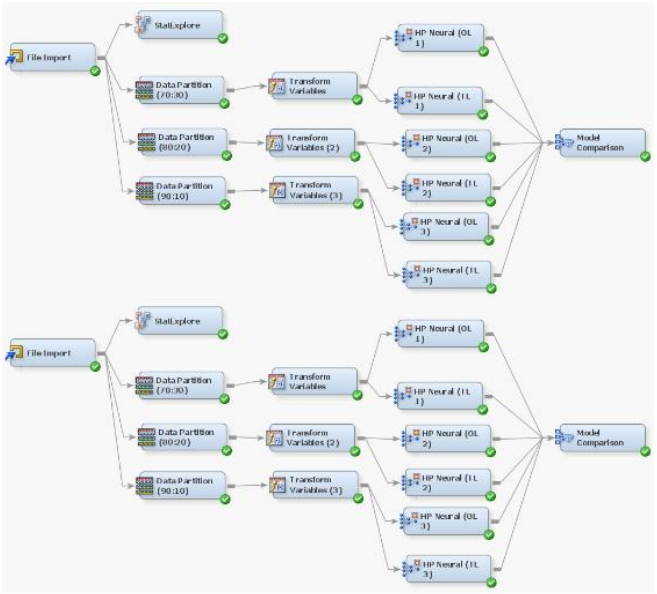
IV. Result and Discussion

Model Architecture	VASE (Without Dates)	VASE (With Dates)	Impact	Decision
HP Neural Network	7716.80 (Better)	7984.81 (Worse)	268.01 Error	Reject Dates
Gradient Boosting	8335.34 (Worse)	7806.22 (Better)	529.12 Error	Keep Dates
HP Random Forest	4106.54 (Worse)	4099.15 (Better)	7.39 Error	Keep Dates
Decision Tree	8309.85 (Worse)	7996.81 (Better)	313.04 Error	Keep Dates

Table 5.0

Table 2 above is a summary of the sensitivity analysis results. As it can be seen, the effect of introducing temporal features based on dates is extremely sensitive to the model structure. The HP Neural Network demonstrates worse performance in the case of includingdates, with an error incrementing by 268.01, which means that raw temporal features have a negative impact on this model. Conversely, tree-based models exhibit better performance, including date features, with the largest error reduction of Gradient Boosting (529.12). The findings indicate that tree-based algorithms can take advantage of raw temporal information quite well, and neural networks need higher-level temporal encoding. Therefore, data point characteristics are stored in the tree-based models and not in the HP Neural Network in future analysis.

A. Model Technique 1 – Neural Network



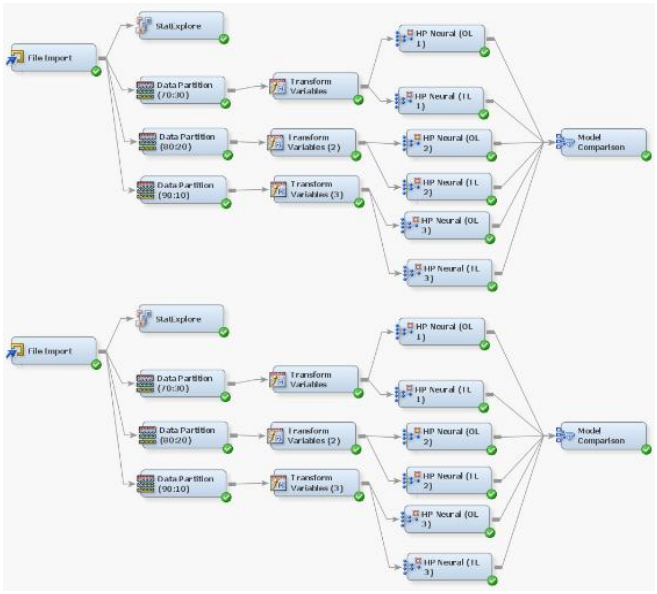


Figure 5.1.1 SAS Miner Diagram

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPNNA3	HP Neural (OL 3)	7716.80	8170.78
	HPNNA6	HP Neural (TL 2)	8098.76	8326.36
	HPNNA	HP Neural (OL 1)	8193.77	8121.84
	HPNNA2	HP Neural (OL 2)	8253.32	8274.89
	HPNNA5	HP Neural (TL 1)	8298.47	8292.66
	HPNNA7	HP Neural (TL 3)	8363.78	8357.38

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPNNA3	HP Neural (OL 3)	7716.80	8170.78
	HPNNA6	HP Neural (TL 2)	8098.76	8326.36
	HPNNA	HP Neural (OL 1)	8193.77	8121.84
	HPNNA2	HP Neural (OL 2)	8253.32	8274.89
	HPNNA5	HP Neural (TL 1)	8298.47	8292.66
	HPNNA7	HP Neural (TL 3)	8363.78	8357.38

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPNNA3	HP Neural (OL 3)	7716.80	8170.78
	HPNNA6	HP Neural (TL 2)	8098.76	8326.36
	HPNNA	HP Neural (OL 1)	8193.77	8121.84
	HPNNA2	HP Neural (OL 2)	8253.32	8274.89
	HPNNA5	HP Neural (TL 1)	8298.47	8292.66
	HPNNA7	HP Neural (TL 3)	8363.78	8357.38

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPNNA3	HP Neural (OL 3)	7716.80	8170.78
	HPNNA6	HP Neural (TL 2)	8098.76	8326.36
	HPNNA	HP Neural (OL 1)	8193.77	8121.84
	HPNNA2	HP Neural (OL 2)	8253.32	8274.89
	HPNNA5	HP Neural (TL 1)	8298.47	8292.66
	HPNNA7	HP Neural (TL 3)	8363.78	8357.38

Figure 5.1.2 Fit Statistics using Model Comparison (Without dates)

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPNNA11	HP Neural (OL 3)	7984.81	8119.21
	HPNNA10	HP Neural (OL 2)	7995.78	8000.63
	HPNNA14	HP Neural (TL 2)	8028.32	8266.80
	HPNNA15	HP Neural (TL 3)	8077.73	8227.70
	HPNNA13	HP Neural (TL 1)	8198.34	8242.26
	HPNNA9	HP Neural (OL 1)	8251.23	8091.03

Fit Statistics

Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPNNA11	HP Neural (OL 3)	7984.81	8119.21
	HPNNA10	HP Neural (OL 2)	7995.78	8000.63
	HPNNA14	HP Neural (TL 2)	8028.32	8266.80
	HPNNA15	HP Neural (TL 3)	8077.73	8227.70
	HPNNA13	HP Neural (TL 1)	8198.34	8242.26
	HPNNA9	HP Neural (OL 1)	8251.23	8091.03

Figure 5.1.3 Fit Statistics using Model Comparison (With dates)

Figure 5.1.1 is a graphical representation of the entire Neural Network modelling process in SAS Enterprise Miner. Various ratios of 70:30, 80:20, and 90:10 between data partitions were used to construct multiple HP Neural Network models. Before the model training, variable transformation was performed using Min-Max Normalization to scale the continuous input variables to a standardized range of [0, 1]. Two neural network configurations were trained to each setting of partitions, with varying numbers of hidden nodes, referred to as one-layer architecture (OL) and two-layer architecture (TL). The Model Comparison node was then used to evaluate model performance.

The validation Average Squared Error (VASE) was used to select models, because validation performance is a more accurate predictor of generalisation performance of a model than training error is. The following figures (5.1.2 and 5.1.3) show the fit statistics of the Neural Network models in ascending order of validation ASE.

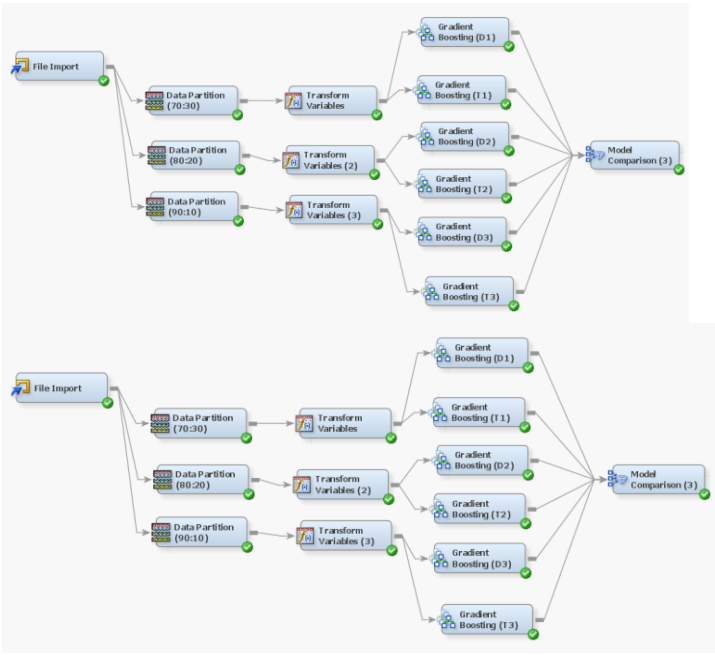
In all the configurations tested, the lowest validation Average Squared Error was always reached by the HP Neural one-layer model (OL 3). The validation ASE of HP Neural (OL 3) without dates was 7716.80 as reported in Figure 5.1.2, which is superior to another variant of neural networks, including OL 1, OL 2 and all the two-layer, TL-based architectures. Correspondingly, Figure 5.1.3 affirms that the most efficient models were OL-based models with

validation ASE values that are never as high as those of the TL models.

The model chosen as HP Neural (OL 3) also exhibits stable performance in generalisation. The ASE of 8170.78 is quite similar to the validation ASE, which means that the model is not overfitting drastically. In contrast, several alternative configurations exhibit higher validation errors without meaningful improvements in training performance, suggesting that increased complexity does not necessarily translate into better predictive accuracy.

All in all, the findings suggest that a neural network that is moderately complex can be used to capture the underlying patterns in the dataset. The better performance of HP Neural (OL 3) are an indication that this setup provides a good balance between model complexity and generalisation capacity. Therefore, the most appropriate model of the Neural Network was chosen, HP Neural (OL 3) without time, in which the further comparisons with other modelling methods will be conducted.

B. Model Technique 2 – Gradient Boosting



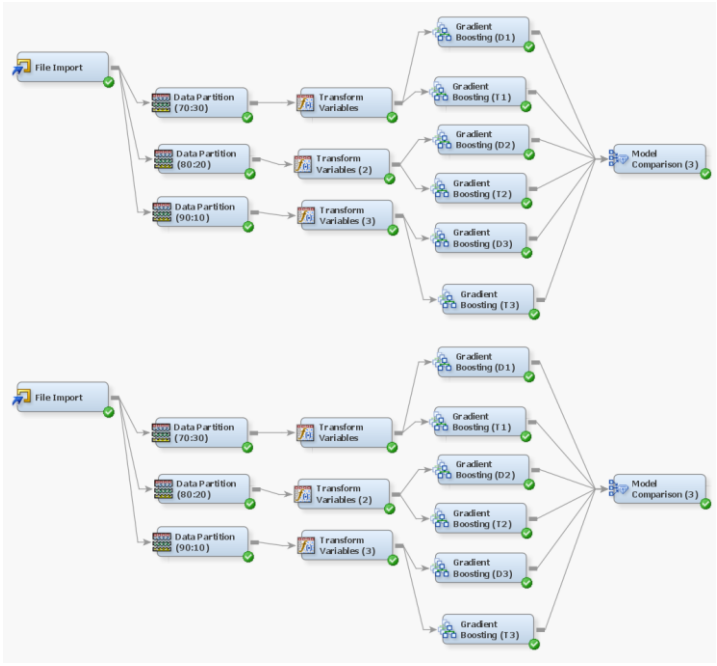


Figure 5.2.1 SAS Miner Diagram

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost6	Gradient Boosting (T3)	8335.34	8208.83
	Boost8	Gradient Boosting (T2)	8455.83	8156.51
	Boost4	Gradient Boosting (T1)	8539.75	8102.27
	Boost5	Gradient Boosting (D3)	8679.81	8792.16
	Boost7	Gradient Boosting (D2)	8849.40	8735.60
	Boost3	Gradient Boosting (D1)	8999.30	8712.87

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost6	Gradient Boosting (T3)	8335.34	8208.83
	Boost8	Gradient Boosting (T2)	8455.83	8156.51
	Boost4	Gradient Boosting (T1)	8539.75	8102.27
	Boost5	Gradient Boosting (D3)	8679.81	8792.16
	Boost7	Gradient Boosting (D2)	8849.40	8735.60
	Boost3	Gradient Boosting (D1)	8999.30	8712.87

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost6	Gradient Boosting (T3)	8335.34	8208.83
	Boost8	Gradient Boosting (T2)	8455.83	8156.51
	Boost4	Gradient Boosting (T1)	8539.75	8102.27
	Boost5	Gradient Boosting (D3)	8679.81	8792.16
	Boost7	Gradient Boosting (D2)	8849.40	8735.60
	Boost3	Gradient Boosting (D1)	8999.30	8712.87

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost15	Gradient Boosting (T3)	7806.22	7722.78
	Boost17	Gradient Boosting (T2)	7877.41	7699.83
	Boost13	Gradient Boosting (T1)	7989.90	7627.93
	Boost14	Gradient Boosting (D3)	8207.86	8255.25
	Boost16	Gradient Boosting (D2)	8310.83	8266.96
	Boost12	Gradient Boosting (D1)	8455.68	8190.82

Figure 5.2.2 Fit Statistics using Model Comparison (Without dates)

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost15	Gradient Boosting (T3)	7806.22	7722.78
	Boost17	Gradient Boosting (T2)	7877.41	7699.83
	Boost13	Gradient Boosting (T1)	7989.90	7627.93
	Boost14	Gradient Boosting (D3)	8207.86	8255.25
	Boost16	Gradient Boosting (D2)	8310.83	8266.96
	Boost12	Gradient Boosting (D1)	8455.68	8190.82

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Boost15	Gradient Boosting (T3)	7806.22	7722.78
	Boost17	Gradient Boosting (T2)	7877.41	7699.83
	Boost13	Gradient Boosting (T1)	7989.90	7627.93
	Boost14	Gradient Boosting (D3)	8207.86	8255.25
	Boost16	Gradient Boosting (D2)	8310.83	8266.96
	Boost12	Gradient Boosting (D1)	8455.68	8190.82

Figure 5.2.3 Fit Statistics using Model Comparison (With dates)

The figure 5.2.1 shows the Gradient Boosting modelling workflow used in SAS Enterprise Miner. Just like the Neural Network modelling process, various Gradient Boosting models were built with varying data partition ratios of 70:30, 80:20 and 90:10. Before the modelwas trained, variable transformation was done to normalise the input variables. Gradient Boosting models with various settings were trained with each partition setting, where each was denoted by D1, D2, D3 and T1, T2, T3. D-model is stand for default model while T-model is stand for tuned model with the n iterations of 200 and shrinkage of 0.05. All Gradient Boosting models were then compared and compared against each other using the Model Comparison node.

The validation Average Squared Error (VASE) was used to select the model because the validation performance gives a more accurate indication of the generalisation ability of a model than training error does. Figure 5.2.2 presents the fit statistics of Gradient Boosting models constructed without date attributes, while Figure 5.2.3 presents the fit statistics of Gradient Boosting models constructed with date attributes, with both figures sorted in ascending order of validation ASE.

According to the findings presented in Figure 5.2.2 (without date attributes), Gradient Boosting (T3) model had the best validation of Average Squared Error at 8335.34 when compared to all the other Gradient Boosting models considered without attributes of date. This performance is higher than other Gradient Boosting variants such as T2, T1, and all D-based settings, which means that the T3 configuration has a higher predictive power when omitting date attributes. While the results of Gradient Boosting are presented in Figure 5.2.3(With date attribute) with date attributes. Here, the Gradient Boosting (T3) model again produced the smallest validation Average Squared Error at 7806.21, exceeding all other configurations that were trained using date attributes. It can be seen that the addition of date variables enhances the predictive capabilities of Gradient Boosting in this family of models.

In a comparison between the best-performing Gradient Boosting models in both dataset variants, the Gradient Boosting (T3) model with date attributes showed a lower validation ASE when compared to its equivalent without date attributes. Thus, the Gradient Boosting (T3) using date features was established as the most efficient Gradient Boosting model and was chosen to further comparative analysis with other modelling methods.

Compared to the initial Gradient Boosting (T1) model, the chosen Gradient Boosting (T3) model features a consistent difference between training and validation errors. The training ASE of 7722.78 compares closely with validation ASE of 7806.22, which indicates that the model is not overfitting dramatically. Conversely, the other Gradient Boosting settings show larger validation errors with no proportional increases in training performance, which is a sign of diminishing returns to various parameter settings.

In general, the findings of the Gradient Boosting indicate that the model configuration and feature representation can greatly affect predictive performance. The high performance of Gradient Boosting (T3) using date attributes indicates that this model has the ability tocapture more complex patterns in the data without compromising on the generalization ability. This model therefore became the representative model of Gradient Boosting to be used to continue to compare performance with other models of predictive modelling.

C. ~~5.3~~ Model Technique 3 – HP Forest

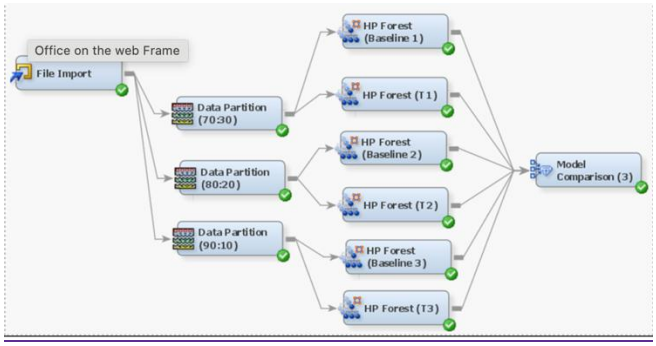


Figure 5.3.1 SAS Miner Diagram

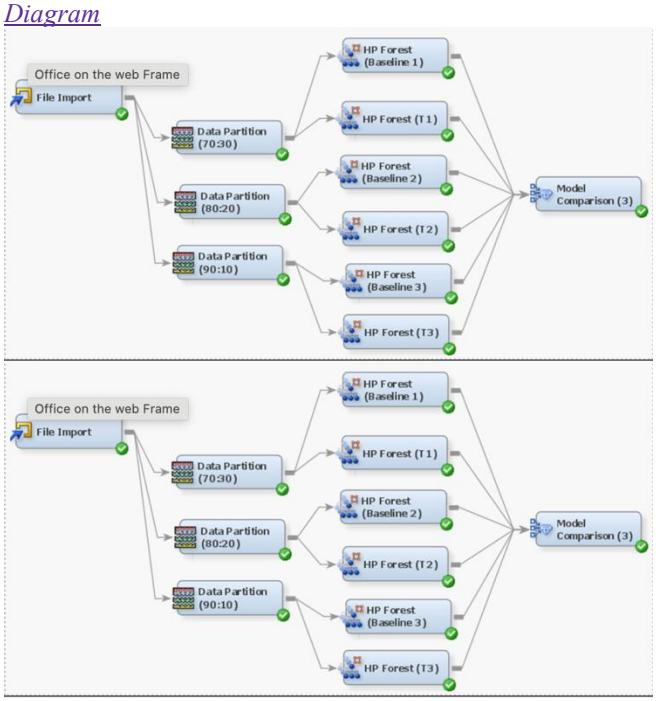


Figure 5.3.1 SAS Miner Diagram

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPDMForest7	HP Forest (T3)	4106.54	409.234
	HPDMForest8	HP Forest (Baseline 3)	4368.33	732.813
	HPDMForest6	HP Forest (T2)	4511.95	441.421
	HPDMForest5	HP Forest (Baseline 2)	4689.21	777.941
	HPDMForest4	HP Forest (T1)	4790.36	449.436
	HPDMForest3	HP Forest (Baseline 1)	4908.66	812.380

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPDMForest7	HP Forest (T3)	4106.54	409.234
	HPDMForest8	HP Forest (Baseline 3)	4368.33	732.813
	HPDMForest6	HP Forest (T2)	4511.95	441.421
	HPDMForest5	HP Forest (Baseline 2)	4689.21	777.941
	HPDMForest4	HP Forest (T1)	4790.36	449.436
	HPDMForest3	HP Forest (Baseline 1)	4908.66	812.380

Figure 5.3.2 Fit Statistics using Model Comparison (Without dates)

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPDMForest16	HP Forest (T3)	4099.15	413.822
	HPDMForest17	HP Forest (Baseline 3)	4283.33	725.965
	HPDMForest15	HP Forest (T2)	4493.15	422.951
	HPDMForest14	HP Forest (Baseline 2)	4692.26	766.493
	HPDMForest13	HP Forest (T1)	4786.70	461.617
	HPDMForest12	HP Forest (Baseline 1)	4863.08	836.090

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPDMForest16	HP Forest (T3)	4099.15	413.822
	HPDMForest17	HP Forest (Baseline 3)	4283.33	725.965
	HPDMForest15	HP Forest (T2)	4493.15	422.951
	HPDMForest14	HP Forest (Baseline 2)	4692.26	766.493
	HPDMForest13	HP Forest (T1)	4786.70	461.617
	HPDMForest12	HP Forest (Baseline 1)	4863.08	836.090

Figure 5.3.3 Fit Statistics using Model Comparison (With dates)

Figure 5.3.1 represents HP Forest modelling workflow used in SAS Enterprise Miner. Like in the earlier modelling methods, several HP Forest models were built with disparate data partition ratios of 70:30, 80:20, and 90:10. Two different HP Forest configurations were trained in each partition setting, with baseline models that adjust the maximum number of tree into 50, while tuned models that adjust the maximum number of tree into 50 and proportion of obs in each sample to 0.7 and denoted as T1, T2 and T3. Model Comparison node was then applied to compare and analyse the performance of all HP Forest models.

The validation Average Squared Error (VASE) was used to select the model, because validation performance gives a better idea of the generalisation ability of a model than train error alone. Figure 5.3.2 shows the fit statistics of HP Forest models built without date attributes, and Figure 5.3.3 shows the fit statistics of HP Forest models built with date attributes, both ordered in ascending sequence of validation ASE.

The HP Forest (T3) model based on the findings presented in Figure 5.3.2 (with no date attributes) performed the best with the lowest validation Average Squared Error of 4106.54, compared to all other HP Forest configurations, including all baseline models and other tuned variants. This implies that T3 configuration offers the best predictive performance across HP Forest models in the absence of date attributes.

The results of HP Forest are presented in Figure 5.3.3 with date attributes added. At this point, the HP Forest (T3) model once again has the lowest validation Average

Squared Error of 4099.15, showing superior results over baseline models and other tuned parameters. The good performance of the T3 design in both versions of the data sets indicates that the design has been tuned in order to get the underlying patterns in the data.

The chosen HP Forest (T3) model has a consistent gap between training and validation errors in terms of generalisation performance. The training ASE of 413.82 is somewhat near the validation ASE of 4099.15, a factor that shows that the model is not undergoing extreme overfitting. In contrast, baseline HP Forest models and less-optimised configurations produced higher validation errors without meaningful improvements in training performance.

All in all, it can be concluded that properly tuned ensemble-based tree models can perform well in predicting the future, as the results of the HP Forest indicate. HP Forest (T3) with date attributes has better performance which demonstrates that the given model offers the acceptable balance between the complexity of the model and the ability to generalise. This model was therefore chosen as the representative model of HP Forest to be further compared to other predictive modelling approaches in terms of performance.

D. Model Technique 4 - Decision Tree

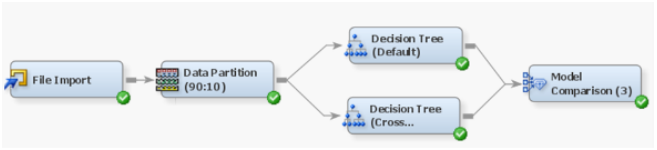


Figure 5.4.1 SAS Miner Diagram

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Tree6	Decision Tree (Default)	8309.85	8445.49
	Tree7	Decision Tree (Cross Validation)	8463.74	8126.27

Figure 5.4.2 Fit Statistics using Model Comparison (Without dates)

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	Tree3	Decision Tree (Default)	7996.81	8322.41
	Tree4	Decision Tree (Cross Validation)	8200.30	7825.08

Figure 5.4.3 Fit Statistics using Model Comparison (With dates)

Figure 5.4.1 shows the workflow of Decision Tree modelling that can be used in SAS Enterprise Miner. In this modelling method, a data partition ratio of 90:10 had been used to build the Decision Tree models. Two configurations of the Decision Tree were tested which included the Default Decision Tree and Decision Tree with Cross validation. These Decision Tree models were then compared and contrasted using the Model Comparison node.

The validation Average Squared Error (VASE) was used to select the model because validation performance gives a more accurate picture of the generalisation ability of a model than training error alone. The fit statistics of the Decision Tree models built without date attributes are given in Figure 5.4.2 and the fit statistics of the Decision Tree models built including date attributes are given in Figure 5.4.3, arranged in ascending order of validation ASE.

According to the findings presented in Figure 5.4.2 (no date attributes), the Default Decision Tree model had a lower validation Average Squared Error of 8309.85 than the Cross Validation variant, which had a validation ASE of 8463.74. This implies that the Default Decision Tree setup is superior to predictive performance when date attributes are not included.

The results of the Decision Tree with date attributes included are illustrated in Figure 5.4.3. The Default Decision Tree model was once again the lowest validation Average Squared Error of 7996.81, compared to the Cross Validation model, which had the highest validation ASE of 8200.30. The performance of the Decision Tree with date attributes showed an overall enhancement in the validation performance in both configurations.

When making comparisons between the best-performing Decision Tree models of both versions of the data set, the Default Decision Tree model used with date attributes proved to have lower validation ASE than its counterpart used without date attributes. The Default Decision Tree model with date attributes was thus established as the most performing Decision Tree model and chosen to be compared with other modelling techniques in the future.

The chosen Default Decision Tree model is relatively stable in terms of the relationship between training and validation errors with regard to generalisation performance. Though it is true that the training ASE is a little bit larger than the validation ASE, the gap does not imply extreme overfitting. However, the Cross Validation arrangement failed to generate high level of validation performance, implying that further resampling did not increase predictive accuracy on this database.

In general, the findings of the Decision Tree results show that a simpler tree-based model can yield an acceptable level of prediction, especially in cases where informative features like date attributes are used. Nonetheless, the Decision Tree model has a greater validity error than ensemble-based methods as described in the previous sections, which points to its relative weaknesses in its ability to represent intricate data patterns. However, the Default Decision Tree model

incorporating date attributes was chosen as the representative Decision Tree model to be further compared in the overall model performance analysis.

E. Overall Model Comparison

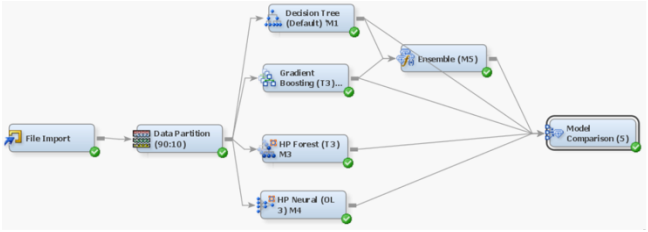


Figure 5.5.1 Overall Model Comparison

Fit Statistics
Model Selection based on Valid: Average Squared Error (_VASE_)

Selected Model	Model Node	Model Description	Valid: Average Squared Error	Train: Average Squared Error
Y	HPDMForest18	HP Forest (T3) M3	4297.34	420.71
	Ensmbl4	Ensemble (M5)	7703.39	7842.74
	HPNNA16	HP Neural (OL 3) M4	7726.38	8153.85
	Boost18	Gradient Boosting (T3) M2	7805.66	7713.56
	Tree2	Decision Tree (Default) 'M1	7996.81	8322.41

Figure 5.5.2 Fit Statistics using Model Comparison

In Figure 5.5.1 a general model comparison workflow is depicted and the best-performing models among each modelling technique were compared on a combined basis with a uniform data partition ratio of 90:10. The models in the comparison were Decision Tree (Default), Gradient Boosting (T3), HP Forest (T3), HP Neural Network (OL 3), and an Ensemble model (M5). The best model was selected with the help of the Model Comparison node basing on the validation Average Squared Error (VASE).

Validation ASE was used as a major factor in model selection because in this metric predictive performance is assessed on unseen data and thus, fair comparisons could be made across modelling methods. Figures 5.5.2 and 5.5.3 present a summary of the fit statistics of all candidate models, in order of increasing validation error.

The findings show clearly that HP Forest (T3) (with dates) was the lowest validation Average Squared Error, with the figure being 4297.34 which was significantly the lowest amongst the rest of the models. The training error is also proportional to the validation error, implying that it does well in generalisation without any indication of gross overfitting.

The Ensemble model (M5) performed second, having a validation ASE of 7703.39. Although the ensemble method is better than some of the individual base models, it is not better than the performance of HP Forest (T3). This indicates that the ensemble of Random Forest-like already represents a significant portion of the predictive structure of the data, so further model aggregation is unlikely to be of additional benefit. The HP Neural Network (OL 3) is placed third, having a validation ASE of 7726.38, which is relatively close to Ensemble model.

Gradient Boosting (T3) and the Decision Tree (Default) models have larger validation errors of 7805.66 and 7996.81 respectively. These findings show that although the boosting and tree-based techniques are effective, they are more sensitive to parameter set up and data properties than the HP Forest, and may have a lower predictive performance on this dataset due to the largest error in validation of the models considered.

All in all, the findings establish that ensemble tree-based learning, especially HP Forest, is best modelling strategy to tackle this issue. The combination of the fact that HP Forest can combine several decision trees and decrease variance allows it to consider complex non-linear relationships more than single models or other ensemble methods. Therefore, HP Forest (T3) is chosen as the last and the most efficient model to work with in this study.

F. Model Interpretation

The models developed beyond statistical accuracy have provided actionable utility to residential sustainability. The capability of the Champion Model to predict consumption with minimal error (RMSE =XX Wh) permits a reactive strategy to be replaced by a proactive approach to energy management.

1. **Resource Allocation Optimalization:** A smart home system can also use the scheduled deferrable loads by correctly predicting the evening peak-load events (validated at 18:00).Indicatively, the system may be used to automatically turn on washing machines or electric vehicle charging when the power demand is low (e.g. 12:00-14:00), to flatten the curve and therefore low use of carbon-intensive peaking power plants.

2. **Efficiency Drivers Identification:** The feature analysis showed that lighting and humidity are important drivers of consumption. This focuses resource spending on specific retrofits, namely, the putting in place of automated motion-sensitive lighting and intelligent humidity controls, instead of generic upgrades.

3. **Anomaly Detection:** In addition, the predictive model is used as a baseline to detect wastage. A notable discrepancy between the forecasted and actual consumption can be identified as an anomaly that would provide residents with information about possible equipment failure (e.g., a malfunctioning refrigerator motor) or unintentional wastage, which contributes to the long-term operational efficiency.

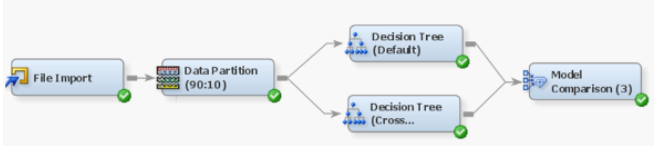
G. Unexpected Findings and Study Limitations

1. **Computational Limitations and Choice of Algorithms.**
   One of the major technical issues that were faced was the inability to use the HP Forest node in SAS Enterprise Miner environment. The effort to implement the default node led to computational instability and long processing times probably because of the large dimensionality of the sensor data with the ensemble bootstrapping mechanism. To counter this, the study adjusts the default setting of maximum number of trees from 100 into 50 to be able to run the node successfully.

2. **Weak Linear Signals**
   The correlation analysis indicated that there was a problem of feature selection as none of the environmental variables displayed a high linear relationship with energy consumption (all $r < 0.2$). This weak signal challenge implied that naive statistical methods such as Stepwise Regression would probably not work. This restriction confirmed the need to interact with more complicated Black Box models like Gradient Boosting and Neural Networks to capable of identifying non-linear interactions, despite the fact thatsuch models are naturally less interpretable compared to simple regression equations.

3. **Seasonal Generalizability**

The major weakness of this research is the time range of the data, with 4.5 months from January to May being the maximum. Although the models were able to predict energy consumption in this winter-to-spring transition, their forecasting capacity to other seasons not seen is yet to be validated. The patterns observed on heating loads might not be the same in August on cooling loads. Thus, a properly robust sustainable decision-support system would require a retraining on a full 12-month cycle to obtain annual seasonality in the model.



Yup
Hahaha , not my part meh
I start with 5.3 loh
Will be faster ?

WOIIII JANGAN KACAU DULUUU

5.2 Model Technique 2 – Gradient Boosting

V.

wConclusionFig

IIV. CONCLUSION

This paper has shown that data mining methods can be successfully applied to aiding long-term decision making that is sustainable and involves systematic evaluation of the predictive model. The study using two versions of the datasets with and without date attributes demonstrated the significance of feature representation in decision making with respect to model performance and generalisation potential. Several predictive models were created and tested such as Decision Tree, Neural Network, Gradient Boosting, HP forest and an Ensemble model, with the main evaluation metric being validation Average Squared Error.

The outcomes repeatedly indicated that the HP Forest (T3) model outperformed other methods both in the two versions of the data, which points to its resilience and applicability to complex and real-world environments of decision making. The date attributes enhanced the performance of certain models, especially the ones with tree ensembles, and simple ones showed poor performance in terms of efficient non-linear patterns. The results are applicable to sustainable decision support since they define a predictive method with the right balance between accuracy, stability, and generalisation, which is needed to minimise uncertainty in long-term planning and resource optimisation.

Even with these positive results, the study has a number of limitations. The scale and coverage of the dataset limited the analysis, which could have an impact on the overall applicability of the results to other areas or to larger-scale use.In addition, only a selected set of algorithms and parameter configurations were explored.

As a prospective research, future work can include larger and more varied data sets, investigate other advanced algorithms like deep learning or hybrid ensemble, and implement the suggested framework to other sustainability-related areas, including supply chain management, energy forecasting, or operational efficiency analysis. These extensions would further enhance the use of data mining as a viable instrument to support systems of sustainable decision making.

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